Data Preparation and Exploratory Data Analysis

Formula 1 Race Intelligence – Part 1 of 4

1. Data Collection

1.1 Data Sources

Source	Description
race_results	One row per driver per race (6,464 records, 17 seasons). Contains season, race_date, race_id, driver, team, finish_position, points, pit_stops, starting_compound, circuit_type, is_wet_race, dnf, and dnf_reason.
driver_nationality	Lookup table giving the nationality of each driver.
circuits (f1_project.xlsx)	Reference table of 29 circuits: circuit_id, circuit_name, country, city, laps, lap_km, type, altitude_m, direction.

1.2 Connecting to Tableau

1. Open Tableau Public Desktop and choose Microsoft Excel under Connect.
2. Add race_results and driver_nationality to the canvas and relate them on the driver field.
3. Check the data in the Data Source tab (6,464 rows expected).
4. Save as a packaged workbook (.twbx) so the data travels with the file.

2. Data Inspection and Validation

Check	How	Expected result
Record count	Count of rows in race_results	6,464
Missing values	Filter each key field for Null	No nulls in driver, team, season, race_id
Data types	Data Source tab	Dates as Date; points, finish_position, pit_stops as Number; country with a geographic role
Duplicates	Compare race_id + driver	One row per driver per race
Name consistency	List distinct team and driver names	No spelling variants of the same team or driver
Flag fields	Check dnf and is_wet_race	Only 0/1 (or True/False) values

3. Data Preparation

3.1 Calculated Fields

Field	Formula	Purpose
Podium	`IF [finish_position] <= 3 THEN 1 ELSE 0 END`	Marks a top-three finish
Win	`IF [finish_position] = 1 THEN 1 ELSE 0 END`	Marks a race win
DNF Rate	`SUM([dnf]) / COUNT([driver])`	Share of entries that did not finish
Reliability %	`(1 - SUM([dnf]) / COUNT([driver])) * 100`	Share of entries that finished
Races	`COUNTD([race_id])`	Number of distinct races
Your Team	Parameter-driven field	Highlights one chosen team against Red Bull Racing and Mercedes

3.2 Other Preparation Steps

- Set country as a geographic role for the host-country map.
- Create a parameter so the user can choose which team is "Your Team".
- Create filters for season, team, driver, wet race, DNF, and DNF reason.
- Keep only the fields needed for the dashboards to improve performance.

4. Exploratory Data Analysis

4.1 Circuits Reference Table

Measure	Finding
Circuits	29 rows, 9 columns, no missing values or duplicates
Circuit types	Permanent 21, Street 5, Street/Park 3
Direction	Clockwise 20, Anti-clockwise 8, Mixed 1
Laps	44 (Spa) to 87 (Bahrain Outer)
Lap length	3.337 km (Monaco) to 7.004 km (Spa)
Altitude	2 m (Miami) to 2,240 m (Mexico)
Countries	24; USA and Italy each have 3 circuits

4.2 Race Results: Questions Explored

Question	Fields used	Worksheet
How do team points change season to season?	season, team, points, finish_position	Team Championship Trajectory
Does the starting compound affect results by circuit type and in wet races?	starting_compound, circuit_type, is_wet_race, points	Tyre Strategy
Which nationalities produce the most points and podiums?	nationality, points, Podium	Talent Pools

Question	Fields used	Worksheet
Who scores the most points, and how do pit stops relate?	driver, season, points, pit_stops	Driver Leaderboard
Which drivers and teams reach the podium most?	driver, team, Podium	Podiums by Driver / Podiums by Team
Which countries host the most races?	country	GP Host Countries
How has reliability changed over time?	season, Reliability %	Reliability Trend by Season
Which failures affect which teams and circuit types?	dnf, dnf_reason, team, circuit_type	Failure Type by Team and Circuit Type

4.3 Findings to Record

After exploring each worksheet, write the main finding here in one sentence with a figure (for example the top driver and points total, the team with the most podiums, the season with the lowest reliability).

5. Issues and Decisions

- `bahrain` and `bahrain_outer` are two layouts of the same venue, so the city and country repeat.
- Decide whether to keep `Street/Park` separate or group it with `Street`.
- Rename the workbook's worksheets so their names match their content, and delete the empty Sheet 12.